

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

Contents

- 1: Purpose..... 2
- 2: Scope..... 2
- 3: Terms and Definitions 2
- 5: Context View..... 5
- 6: Physical View..... 8
- 7: Logical View..... 10
- 8: Manufacturing /Service View..... 12
 - a. Device Provisioning 12
 - b. Diagnostics and Debugging 12
- 9: Record History 13
- 10: Record Review and Approval 13

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

1: Purpose

The purpose of this document is to describe the top-level architecture for the software components developed by Empyrean Medical Systems for the Voxel Ray GentleBeam system, known as Project Heracles.

This document presents the software components from multiple points of view, describing the system context and logical software items, including SOUP components, developer configurations, and other viewpoints.

2: Scope

This document describes the software components for the Voxel Ray GentleBeam system, known as Project Heracles, at a high-level. The document does not address requirements, electro-mechanical designs, detailed software designs, or verification protocols (all of which other project documents cover).

*The term Software relates to both software and firmware subsystems.

3: Terms and Definitions

Term	Definition
Architecture	Organizational structure of a system or component.
CM	Configuration Management
Configuration Item (CI)	Entity that can be uniquely identified at a given reference point.
Design History File (DHF)	A compilation of specific records that describe the design history of a product.
SW	Software
OTS	Off-the-shelf
Process	Any activity that receives inputs and converts them into outputs.
Product	Components, manufacturing materials, in-process devices, finished devices, and returned devices.
PRS	Product Requirements Specification
Regulatory Requirement	A statutory requirement delineated within local, state, federal, or international regulations, applicable to the QMS, and employed for the manufacture of medical devices.
Safety	Freedom from unacceptable risk.

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

Safety Defect	An anomaly indicating a product malfunction that may affect the safe operation of the product and may pose harm to patients and/or users of the product.
SBOM	Software Bill of Materials
Software Item	Any identifiable part of a computer program, i.e., source code, object code, control code, control data, or a collection of these items.
Software System	Integrated collection of software items organized to accomplish a specific function or set of functions.
Software Unit	Software item that is not subdivided into other items (defined by the manufacturer).
SOP	Standard Operating Procedure
Software of Unknown Provenance (SOUP)	Software item that is already developed and generally available and that has not been developed for the purpose of being incorporated into the medical device (also known as “off-the-shelf software”) or software item previously developed for which adequate records of the development processes are not available.
Specification	Any requirement with which a product, process, service, or other activity must conform.
RTT	Radiation Therapy Technician
C&C	Command and Control
DAL	Data access Layer

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

4: References

4.1: Empyrean Medical Systems References

Document Identifier	Document Title
DHF-HRC-0001, Rev. 2	Heracles SDP
DHF-HRC-0002	VoxelRay PRS
DHF-HRC-0007, Rev. 3	Heracles External GUI SRS
DHF-HRC-0009, Rev. 3	Heracles Record and Verify SRS
DHF-HRC-0005, Rev. 3	Heracles Internal GUI SRS

4.2: Standard and Regulatory References

#	Standard
1.	IEC 62304 Edition 1.1 2015-06, recognition #13-79
2.	ISO 14971 Medical devices — Application of risk management to medical devices Third edition 2019-12
3.	21 CFR Part 820 Subpart 820.30 - FDA Quality System Regulation
4.	ISO 13485:2016 Clause 7.3 - Medical devices – Quality management systems – Requirements for regulatory purposes

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

5: Context View

The Heracles (AKA GentleBeam) software consists of the following: Heracles Internal SW, Heracles External SW, and Record and Verify SW.

These software components provide the operator with interfaces that enable the delivery of superficial radiation therapy to patients using the Heracles system.

The interfaces will include managing patients' demographics, acquiring images, and control and monitoring the delivery of the radiation treatment.

The diagram below depicts the Context View of the software, illustrating the interaction of various components, system FW, and Image acquisition.

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)				
Document Number:	DHF-HRC-0004	Revision:	2		
Date:	16-OCT-2025	Document Type:	Record		

Figure 1: Context View Diagram

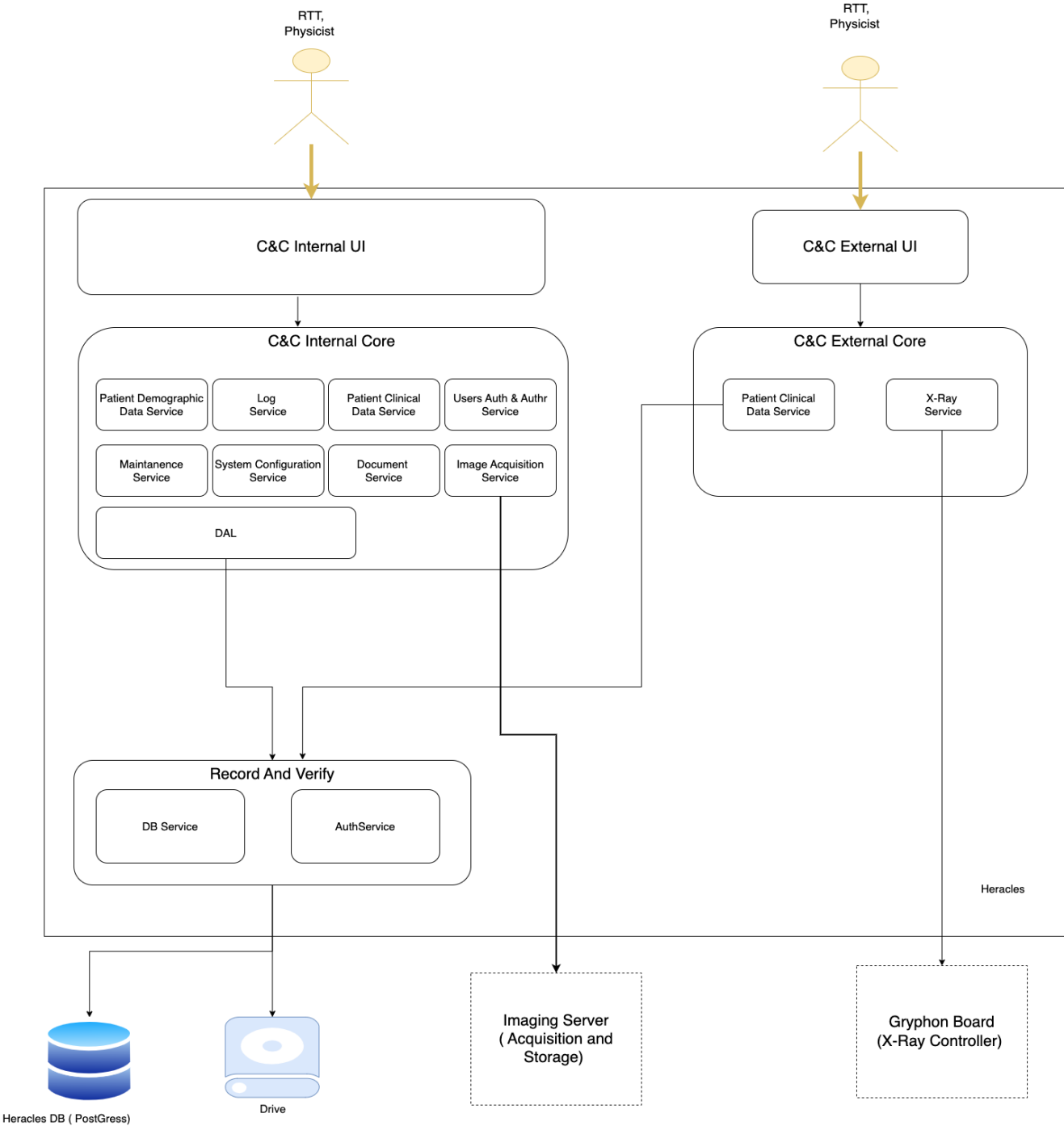


Figure 1: Context View Diagram

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

SW Item	Description
Patient Demographic Data Service	Manages patient demographics data. Enables RTTs to create, view, and update patient details i.e., name, DOB, SSN (or other form of ID) address, etc.
Patient Clinical Data Service	Manages patient clinical related information i.e., diagnosis (site name), prescriptions, treatment plans, and treatments history.
Image Acquisition Service	Allows acquiring medical images using the imaging probe integrated within the Heracles device.
Maintenance Service	Provides the ability to perform system maintenance tasks such as quality checks and daily warmups.
System Configuration Service	Provides the ability to program the system's calibration values,
Users Authorization and Authentication Service	Manages user logins and role-based access. This module ensures that only legitimate users gain access to specific resources within the system. It verifies the identity of a user (authentication) and determines the resources a user is allowed to access and the operations they can perform (authorization).
Document Service	Manage patient clinical related documentation like photos, reports, letters of consent etc.'
Log Service	Maintains the system logs and errors sequence created by the system.
DAL Service	Data access layer. Interface with the record and verify component.
X-ray Service	Responsible for the initiating, terminating, and monitoring the radiation delivery of the selected patient's treatment plan. provides
DB Service	Provides API for creating, retrieving, and updating data.
Auth Service	Provides authorization and authentication services.

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)				
Document Number:	DHF-HRC-0004	Revision:	2		
Date:	16-OCT-2025	Document Type:	Record		

6: Physical View

Physical View maps software subsystems onto physical subcomponents. The system FW and image acquisition components are marked as green boxes, as their architecture is out of scope of this document.

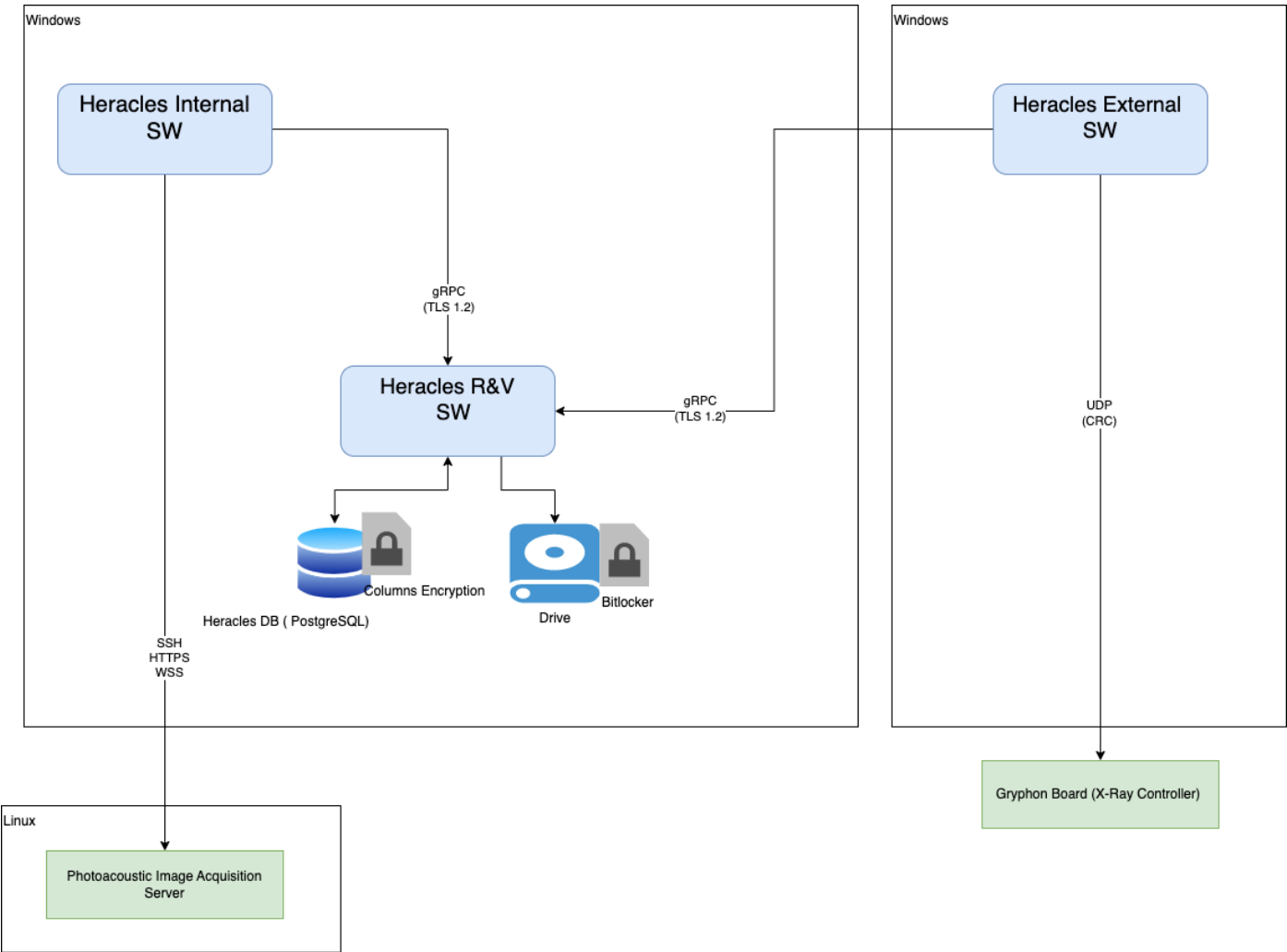


Figure 2: Physical View Diagram

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

Physical View Diagram – Software Items			
<i>Element</i>	<i>Description</i>	<i>Processor</i>	<i>Implementation Language</i>
Heracles C&C Internal software	Addresses functionalities related to patient imaging, demographic data management, clinical data management, treatment data management, image acquisition, user access, clinical workflows, and system setup and services.	Windows	.NET C#
Heracles C&C External software	Used to activate and monitor the radiation for the treated patient and to provide tools for RTT and physicists to perform quality checks, system calibration, and service maintenance. This software should run on a console outside the treatment room. This is to provide the operator with a safe work environment.	Windows	.NET C#
Heracles Record and Verify	Provides APIS to save, update and retrieve data from the database.	Windows	Node JS

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

7: Logical View

Logical View describes the software items, including SOUP, for each subsystem.

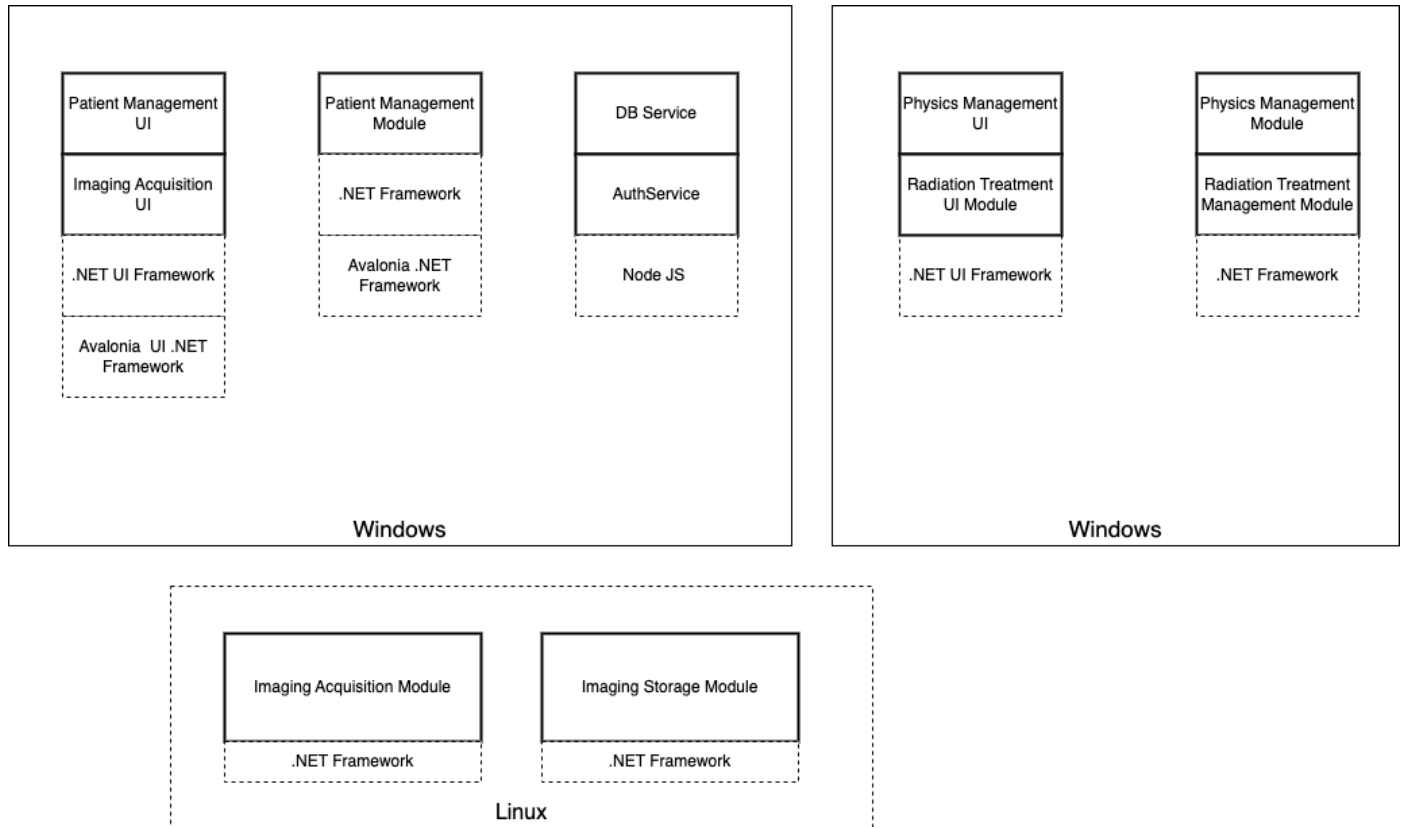


Figure 4: Logical View Diagram

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
Document Number:	DHF-HRC-0004	Revision:	2	
Date:	16-OCT-2025	Document Type:	Record	

The following table lists all the SOUP packages used for significant functionality. The full list of SOUP packages and versions present is found in the Software Bill of Materials (SBOM).

<i>SOUP</i>	<i>Developer</i>	<i>Version</i>	<i>Release Date</i>	<i>Purpose</i>
.NET Framework		8.0		Programs written for .NET Framework execute in a software environment named the CLR (Common Language Runtime)
Prisma Client		5.22.0		Postgres DB communication

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
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Date:	16-OCT-2025	Document Type:	Record	

8: Manufacturing /Service View

Empyrean Medical Systems Manufacturing and Service personnel are provided with tools and data that can be used to access a device's maintenance interface.

1. Maintenance Operations

a. Device Provisioning

- i. Assigning unique identifiers such as **serial numbers**, **software versions**, and **firmware details** during manufacturing or deployment.
- ii. Configuring default settings for integration into the clinical environment.

b. Diagnostics and Debugging

- i. Logging errors and warnings into the system for troubleshooting.
- ii. Providing **debugging tools** to engineers, such as diagnostic UI modules.

c. Event detection and logging

- i. Logging events using **Sysmon** service

2. Updates and Patches

- a. The device will receive at least 1 yearly update during preventative maintenance. The device will receive updates and security patches at any point in time once one will be necessary.
- b. All updates will be performed by a trained service engineer using physical USB drive to deliver the updates to the device. The updates will be executed by an administrative user.
- c. Patching the **application components** to address new features, security vulnerabilities, or performance improvements.
- d. Updating dependent **SOUP libraries** when necessary.

Document Title:	Software Architecture Description for GentleBeam System (Heracles Project)			
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Date:	16-OCT-2025	Document Type:	Record	

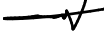
9: Record History

Table 1: Record History

Revision	Rev Date	Author	Description of Changes
1	10-DEC-2024	Limor Erez	Initial Release
2	16-OCT-2025	Limor Erez	New Features: User Authorization, Virtual Keyboard. Enhancements: UI Usability changes. Robotic Arm Control has been removed (only hand guided movement is enabled) Imaging and cybersecurity controls were added.

10: Record Review and Approval

Author	Author Function	Signature and Date
Limor Erez	Director, Software and Products	 Electronically signed by: Limor Erez Reason: I am the author Date: Nov 1, 2025 18:48:13 GMT+2

Reviewer/Approver(s)	Function	Signature and Date
Yonatan Vainer	CTO / PMO	 Electronically signed by: Yonatan Vainer Reason: I am the approver Date: Nov 1, 2025 18:49:57 GMT+2
Adi Fishman	Software & Systems QA/RA Engineer	 Electronically signed by: Adi Fishman Reason: I am the approver Date: Nov 1, 2025 12:51:06 EDT

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